Dhana Karthikeya Ventrapragada

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Education

University of California, Berkeley **Expected Dec 2027**

B.A. in Data Science and Applied Mathematics — GPA: 3.85/4.00 *Berkeley, CA*

- **Relevant Coursework:** Principles & Techniques of Data Science · Probability for Data Science · Data Structures · *In progress:* Introduction to Machine Learning · Efficient Algorithms & Intractable Problems · Data Engineering

Technical Skills

Languages: Python, SQL, Java

ML: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, SentenceTransformers, Embeddings, Cross-Validation

Data: pandas, NumPy, GeoPandas, Matplotlib, PostgreSQL, Jupyter

Engineering: FastAPI, Gemini API, Pydantic, AWS S3, ffmpeg, pytest, Git

Experience

IDX Exchange | Data Science Intern | Remote **Jun 2026 – Aug 2026**

- Developed a model that predicts California home sale prices, typically within **8.6%** of the actual price, using **71K** single-family sales cleaned from **143K** raw records.
- Benchmarked four model types and selected a tuned XGBoost (R^2 of **0.88**), finding that deeper trees improved accuracy more than a lower learning rate.
- Resolved a data leakage bug that had invalidated early results, and nearly halved a simpler model's error (from **29%** to **15%**) by engineering a school district feature.

Guang Labs | Engineering Intern | Remote **May 2026 – Aug 2026**

- Engineered a service for an AI video startup that classifies a user's image or video by content type (sports, gaming, talking, music) using Google's Gemini, retrying uncertain or malformed responses on different clips, up to **10** attempts.
- Validated the service end to end with real video clips after code review and expanded the automated test suite from 38 to **86** tests.
- Developed an in-app notification bell that alerts users when their video finishes processing or fails.

Fung Institute | Fung Fellow, Health + Innovation | Berkeley, CA **Aug 2026 – Present**

- Selected for Berkeley's industry-partner fellowship, placed on The After Collective project supporting wildfire survivors.
- Interviewed wildfire survivors about the tools and information they use, and organize the team's documentation.

Projects

Hindsight (Cal Hacks 13.0) | *Claude, FastAPI, React, Apify, Factory Droid* GitHub

- Co-designed a transfer planner for California community college students that shows which classes count toward their major using official ASSIST agreements, citing a source for every answer.
- Shaped the product idea, features, and vision, and gave feedback that improved the interface and added more useful features.
- Grounded the app in real data so nothing is made up: credits are matched by code against official records, and campus opportunities are scraped from real pages (Apify, run by a Factory Droid agent) with source links.

SightLine (Open Projects, ML team) | *PyTorch, OpenCV, in progress* GitHub

- Developing an app that turns a day of Meta Ray-Ban smart glasses footage into a **60-second** highlight video; on the ML team, which splits footage into clips and uses audio and visual models to pick the best moments.

Movie Genre Classification | *Python, scikit-learn, SentenceTransformers* GitHub

- Built a model that reads a movie's title and plot summary and predicts all its genres, trained on **42K+** films across 20+ genres.
- Raised the overall score (Macro F1) from 0.48 to **0.66** by giving each genre its own cutoff, and found a simple model beat complex ones while training **12**× faster.

Leadership & Involvement

UC Berkeley CDSS | Reader/Tutor, CDSS 198 | Berkeley, CA **Aug 2026 – Present**

- Run exam reviews for intro CS and data science classes for transfer students, and give resume feedback.

UC Berkeley EECS | Course Staff (UCS1), CS 61A | Berkeley, CA **Jun 2026 – Aug 2026**

- Taught **4** weekly exam-prep sections, held office hours, helped write midterm and final questions, and led final review sessions.